

Math 161

The Heine-Borel theorem

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1 Cauchy sequences

Our proof of the Heine-Borel theorem will use convergence of sequences, a topic which won't be officially covered until 162. We'll quickly treat the material needed for Heine-Borel here.

Definition 1.1. A *sequence* of real numbers is a map $\mathbb{N} \rightarrow \mathbb{R}$, which amounts to an infinite list

$$x_1, x_2, x_3, \dots \in \mathbb{R}.$$

We usually denote a sequence by $\{x_n\}_{n \in \mathbb{N}}$ or simply $\{x_n\}$. We say that the sequence $\{x_n\}$ *converges* to $L \in \mathbb{R}$ if for any $\epsilon > 0$, there exists $N > 0$ such that $n \geq N$ implies that $|x_n - L| < \epsilon$. In that case we write

$$\lim_n x_n = L.$$

If $\{x_n\}$ converges to some $L \in \mathbb{R}$, we say that $\{x_n\}$ is *convergent*.

The following is an important fact about sequences which amounts to an alternative characterization of closed sets.

Proposition 1.2. A set $A \subset \mathbb{R}$ is closed if and only if for any convergent sequence $\{x_n\}$ such that $x_n \in A$ for all $n \in \mathbb{N}$, we have $\lim_n x_n \in A$.

Proof. First assume that A is closed, and let $\{x_n\}$ be a convergent sequence such that $x_n \in A$ for all $n \in \mathbb{N}$. Suppose for the sake of contradiction that

$$L := \lim_n x_n \notin A.$$

Since A is closed, it follows that there exists $\epsilon > 0$ such that $(L - \epsilon, L + \epsilon) \cap A = \emptyset$. On the other hand, there exists $N > 0$ such that $n \geq N$ implies that $|x_n - L| < \epsilon$. In particular $x_N \in (L - \epsilon, L + \epsilon) \cap A$, a contradiction.

Now we prove the converse, which will not be used below but is included for completeness. Fix $x \in \mathbb{R} \setminus A$, and for any $n \in \mathbb{N}$, consider the open interval $(x - \frac{1}{n}, x + \frac{1}{n})$. If

$$(x - \frac{1}{n}, x + \frac{1}{n}) \cap A = \emptyset$$

for some $n \in \mathbb{N}$ then it follows that A is closed, so assume otherwise for the sake of contradiction. Then for any $n \in \mathbb{N}$ we can choose $x_n \in (x - \frac{1}{n}, x + \frac{1}{n}) \cap A$. Note that $\lim_n x_n = x$, since for any $\epsilon > 0$ we can choose $N \in \mathbb{N}$ such that $\frac{1}{N} < \epsilon$, and then $n \geq N$ implies that

$$|x_n - L| < \frac{1}{n} \leq \frac{1}{N} < \epsilon.$$

Now our hypothesis on A implies that $x = \lim_n x_n \in A$, contrary to assumption.

□

The following is a useful auxiliary notion.

Definition 1.3. A sequence $\{x_n\}$ is called *Cauchy* if for any $\epsilon > 0$ there exists $N > 0$ such that $m, n \geq N$ implies that $|x_m - x_n| < \epsilon$.

The condition of being Cauchy means that eventually the terms of the sequence get arbitrarily close together. As the next proposition shows, this condition allows us to recognize a convergent sequence without identifying its limit.

Proposition 1.4. *A sequence is Cauchy if and only if it converges.*

We will need the following lemma.

Lemma 1.5. *A Cauchy sequence is bounded.*

Proof. Suppose that $\{x_n\}$ is Cauchy and unbounded below. Since $\{x_n\}$ is Cauchy, there exists $M > 0$ such that $m, n \geq M$ implies that $|x_m - x_n| < 1$. On the other hand, since $\{x_n\}$ is unbounded below there exists $N \geq M$ such that $x_N < x_M - 1$, or equivalently $x_M - x_N > 1$. But $M, N \geq M$ implies that $|x_M - x_N| < 1$, a contradiction.

If $\{x_n\}$ is Cauchy and unbounded above, apply the above argument to the sequence $\{-x_n\}$, which is clearly Cauchy and unbounded below, to obtain a contradiction. □

Proof of Proposition 1.4. First, we show that a convergent sequence is Cauchy (this will not be used below, but is included for completeness). If $\lim x_n = L$, then for any $\epsilon > 0$ there exists $N > 0$ such that $n \geq N$ implies that $|x_n - L| < \frac{\epsilon}{2}$. Then $m, n \geq N$ implies that

$$|x_m - x_n| \leq |x_m - L| + |L - x_n| = |x_m - L| + |x_n - L| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

For the converse, given a Cauchy sequence $\{x_n\}$, consider the following set:

$$A := \{x \in \mathbb{R} \mid \text{there exists } N > 0 \text{ such that } n \geq N \text{ implies that } x_n \geq x\}.$$

First, we claim that A is nonempty. If $A = \emptyset$ then for any $x \in \mathbb{R}$ and $N > 0$ there exists $n \geq N$ such that $x_n < x$, i.e. the sequence $\{x_n\}$ is unbounded below. By the lemma, this is impossible.

Next we claim that A is bounded above. If not, then for any $x \in \mathbb{R}$ there exists $y \in A$ with $y \geq x$, and hence there exists $N > 0$ such that $n \geq N$ implies that $x_n \geq y \geq x$. That is, the sequence $\{x_n\}$ is unbounded above. Again, this contradicts the lemma.

Thus the least upper bound property implies that $L := \sup A$ exists. We claim that $\lim x_n = L$. Given $\epsilon > 0$, we must produce $N > 0$ such that $n \geq N$ implies that $|x_n - L| < \epsilon$. First, since $\{x_n\}$ is Cauchy, there exists $N_1 > 0$ such that $m, n \geq N_1$ implies that $|x_m - x_n| < \frac{\epsilon}{2}$. Since $L = \sup A$, there exists $x \in A$ such that $L - \frac{\epsilon}{2} < x \leq L$ (otherwise $L - \frac{\epsilon}{2}$ would be an upper bound for A , contradicting the fact that L is the least upper bound). By the definition of A , there exists $N_2 > 0$ such that $n \geq N_2$ implies that $x_n \geq x$. On the other hand, since $L + \frac{\epsilon}{2} > L = \sup A$ we have $L + \frac{\epsilon}{2} \notin A$, and hence there exists $N \geq N_1, N_2$ such that $x_N < L + \frac{\epsilon}{2}$. Note that since $N \geq N_2$, we also have

$$x_N \geq x > L - \frac{\epsilon}{2}$$

and hence $|x_N - L| < \frac{\epsilon}{2}$. Finally, if $n \geq N$ then we have

$$|x_n - L| \leq |x_n - x_N| + |x_N - L| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon,$$

where the second inequality used the fact that $n, N \geq N_1$. □

2 The theorem

We recall the statement of the theorem.

Theorem 2.1 (Heine-Borel theorem). *A set $A \subset \mathbb{R}$ is compact if and only if A is closed and bounded.*

The proof will be broken down into several lemmas.

Lemma 2.2. *If $A \subset \mathbb{R}$ is compact, then A is closed.*

Proof. Choose $x \in \mathbb{R} \setminus A$, so we must show that there exists $\epsilon > 0$ such that $(x - \epsilon, x + \epsilon) \cap A = \emptyset$. For any $y \in A$, let $\epsilon_y := \frac{1}{2}|x - y| > 0$ and $U_y := (y - \epsilon_y, y + \epsilon_y)$. Then $\{U_y\}_{y \in A}$ is an open covering of A , and since A was assumed compact there exists a finite subset $A_0 \subset A$ such that $\{U_y\}_{y \in A_0}$ is an open covering of A .

Put

$$\epsilon := \min_{y \in A_0} (\epsilon_y),$$

which makes sense because A_0 is finite. By the definition of ϵ_y , we have $(x - \epsilon, x + \epsilon) \cap U_y = \emptyset$ for any $y \in A$, and in particular for any $y \in A_0$. Since

$$(x - \epsilon, x + \epsilon) \subset (x - \epsilon_y, x + \epsilon_y)$$

for all $y \in A_0$, it follows that $(x - \epsilon, x + \epsilon) \cap U_y = \emptyset$ for all $y \in A_0$. But $A \subset \bigcup_{y \in A_0} U_y$, so we obtain $(x - \epsilon, x + \epsilon) \cap A = \emptyset$ as desired. $\square$

Lemma 2.3. *If $A \subset \mathbb{R}$ is compact, then A is bounded.*

Proof. Consider $\{(-n, n)\}_{n \in \mathbb{N}}$, which is an open covering of $\mathbb{R}$ and hence of A . Since A is compact, there exists a finite subset $I_0 \subset \mathbb{N}$ such that $\{(-n, n)\}_{n \in I_0}$ is an open covering of A . Let $N := \max(I_0)$, so we obtain

$$A \subset \bigcup_{n \in I_0} (-n, n) = (-N, N)$$

and hence A is bounded. $\square$

The following lemma is the most nontrivial input into the theorem, and it is here where sequences are needed.

Lemma 2.4. *For any $a, b \in \mathbb{R}$, the interval $[a, b]$ is compact.*

Proof. If $a > b$ then $[a, b] = \emptyset$, and if $a = b$ then $[a, b] = \{a\}$. In both of these cases it is trivial to show compactness, so we assume that $a < b$.

Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) := a + (b - a)x$. Then f is a polynomial and in particular continuous, and we have $f([0, 1]) = [a, b]$. Thus it suffices to show that $[0, 1]$ is compact, since the continuous image of a compact set is compact.

Suppose that $\{U_i\}_{i \in I}$ is an open covering of $[0, 1]$. We assume for the purpose of contradiction that this open covering admits no finite subcovering. Subdivide the interval in half:

$$[0, 1] = [0, \frac{1}{2}] \cup [\frac{1}{2}, 1].$$

Now $\{U_i\}_{i \in I}$ is an open covering of both $[0, \frac{1}{2}]$ and $[\frac{1}{2}, 1]$, since each is contained in $[0, 1]$. It follows that either $[0, \frac{1}{2}]$ or $[\frac{1}{2}, 1]$ does not admit a finite subcovering: if both did admit a finite subcovering, then we could take the union of both of these finite subcoverings to obtain a finite subcovering of $[0, 1]$, contrary to assumption. Say $[\frac{i}{2}, \frac{i+1}{2}]$ does not admit a finite subcovering where $i = 0$ or 1 , and choose an element $x_1 \in [\frac{i}{2}, \frac{i+1}{2}]$.

We continue this process inductively. Suppose that in the previous step, the subinterval $[\frac{i}{2^n}, \frac{i+1}{2^n}]$ (here $0 \leq i < 2^n$) did not admit a finite subcovering and we chose $x_n \in [\frac{i}{2^n}, \frac{i+1}{2^n}]$. Now subdivide this interval in half:

$$[\frac{i}{2^n}, \frac{i+1}{2^n}] = [\frac{2i}{2^{n+1}}, \frac{2i+1}{2^{n+1}}] \cup [\frac{2i+1}{2^{n+1}}, \frac{2i+2}{2^{n+1}}].$$

By the same reasoning as above, either $[\frac{2i}{2^{n+1}}, \frac{2i+1}{2^{n+1}}]$ or $[\frac{2i+1}{2^{n+1}}, \frac{2i+2}{2^{n+1}}]$ does not admit a finite subcovering: say $[\frac{j}{2^{n+1}}, \frac{j+1}{2^{n+1}}]$ does not admit a finite subcovering where $j = 2i$ or $2i+1$. We choose $x_{n+1} \in [\frac{j}{2^{n+1}}, \frac{j+1}{2^{n+1}}]$.

In this way we obtain a sequence $\{x_n\}$ in $[0, 1]$, which we claim is Cauchy. Namely, for any $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that $\frac{1}{2^N} < \epsilon$. If $m, n \geq N$, then we have $x_m, x_n \in [\frac{i}{2^N}, \frac{i+1}{2^N}]$ for some $0 \leq i < 2^N$ and hence

$$|x_m - x_n| \leq \frac{1}{2^N} < \epsilon.$$

Thus Proposition 1.4 implies that $\{x_n\}$ converges. Letting $L := \lim_n x_n$, Proposition 1.2 implies that $L \in [0, 1]$ because $[0, 1]$ is closed. Since $\{U_i\}_{i \in I}$ is an open covering of $[0, 1]$, we have $L \in U_{i_0}$ for some $i_0 \in I$. Because U_{i_0} is open, there exists $\epsilon > 0$ such that $(L - \epsilon, L + \epsilon) \subset U_{i_0}$. Let $N \in \mathbb{N}$ be such that $\frac{1}{2^N} < \epsilon$, and let $0 \leq i < 2^N$ be the index chosen so that $[\frac{i}{2^N}, \frac{i+1}{2^N}]$ admits no finite subcovering and $x_N \in [\frac{i}{2^N}, \frac{i+1}{2^N}]$. Since $x_n \in [\frac{i}{2^N}, \frac{i+1}{2^N}]$ for all $n \geq N$, another application of Proposition 1.2 implies that $L \in [\frac{i}{2^N}, \frac{i+1}{2^N}]$. But then $\frac{1}{2^N} < \epsilon$ implies that

$$[\frac{i}{2^N}, \frac{i+1}{2^N}] \subset (L - \epsilon, L + \epsilon) \subset U_{i_0}.$$

Thus the singleton $\{U_{i_0}\}$ is an open covering of $[\frac{i}{2^N}, \frac{i+1}{2^N}]$, and in particular is a finite subcovering of this interval, contradicting our assumption. □

Lemma 2.5. *Suppose that $A \subset \mathbb{R}$ is compact and $B \subset \mathbb{R}$ is closed. If $B \subset A$, then B is compact.*

Proof. Let $\{U_i\}_{i \in I}$ be an open covering of B . Since B is closed, the set $\mathbb{R} \setminus B$ is open, and $B \subset \bigcup_{i \in I} U_i$ implies that

$$(\mathbb{R} \setminus B) \cup \bigcup_{i \in I} U_i = \mathbb{R}.$$

In particular, the collection $\{U_i\}_{i \in I} \cup \{\mathbb{R} \setminus B\}$ is an open covering of $\mathbb{R}$, and hence of A . Since A is compact, there exists a finite subset $I_0 \subset I$ such that $\{U_i\}_{i \in I_0} \cup \{\mathbb{R} \setminus B\}$ is an open covering of A , i.e.

$$A \subset (\mathbb{R} \setminus B) \cup \bigcup_{i \in I_0} U_i.$$

It follows from this that

$$B \subset \bigcup_{i \in I_0} U_i,$$

and hence $\{U_i\}_{i \in I_0}$ is a finite subcovering of B as needed. □

The proof of the theorem is now just a matter of combining these lemmas.

Proof of the Heine-Borel theorem. If A is compact, then Lemmas 2.2 and 2.3 precisely say that A is closed and bounded.

For the converse, suppose that A is closed and bounded. Since A is bounded, we have $A \subset [-N, N]$ for some $N > 0$. By Lemma 2.4 tells us that $[-N, N]$ is compact, and since A is closed and $A \subset [-N, N]$, Lemma 2.5 implies that A is compact. □